

QMWS Spring 2026: Introduction to Python for Data Analysis

Day One: Foundations

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Section 1

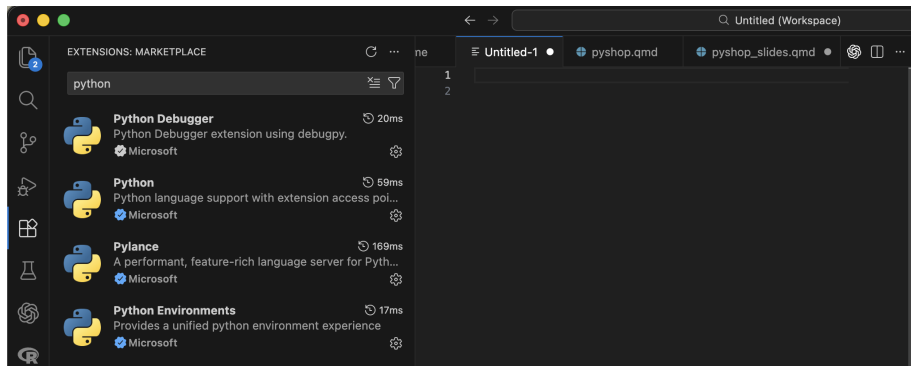
Installation

Visual Studio (VS) Code

- Installation link: <https://code.visualstudio.com/download>
- Integrated Development Environment (IDE)
- Can use it to code in many different languages
- Commonly used in both academia and industry

Visual Studio (VS) Code

- Once installed, go to the “Extensions” tab and search “Python”
- Install the below extensions:



- A tool for configuring Python **environments**
 - More often than not, you need to install **packages** that allow you to code more easily/efficiently
 - Allows for version control
 - Issues can arise if environments are not used, for example:
 - If you install two packages that conflict with one another
 - If you install a package that requires a specific version of Python
 - Anaconda allows these issues to be isolated to a specific environment, rather than affecting your whole Python installation
 - Especially useful if you have more than one project going on

- Installation link: <https://www.anaconda.com/download>
- The simplest way to use Anaconda is via the **Navigator**
- To install:
 - 1 Make an account
 - 2 Install the full distribution via the “Graphical Installer”

- Installation link: <https://www.anaconda.com/download>
- On Mac:



Choose Your Download

Windows

Mac

Linux

Anaconda Distribution

Complete package with 8,000+ libraries, Jupyter, JupyterLab, and Spyder IDE. Everything you need for data science.

↓ 64-Bit (Apple silicon)
Graphical Installer

↓ 64-Bit (Apple silicon)
Command Line Installer

Miniconda

Minimal installer with just Python, Conda, and essential dependencies. Install only what you need.

↓ 64-Bit (Apple silicon)
Graphical Installer

↓ 64-Bit (Apple silicon)
Command Line Installer

↓ 64-Bit (Intel chip) Graphical
Installer

↓ 64-Bit (Intel chip) Command

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 [Windows 64-Bit Graphical Installer](#)

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Section 2

Setting Up a Conda Environment

Section 3

The Basics

- In programming, a variable is a space in computer memory where values may be stored